

SARTHAK CHAWATHE

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EXPERIENCE

AI/ML Engineer | Mayo Clinic

Mar 2025 – Present

Healthcare AI, LLM Evaluation, Workflow Automation, Product Analytics, Vertex AI, BigQuery, HIPAA

Remote, USA

- Led prompt engineering, fine-tuning, and evaluation of **Gemini-based LLM workflows** for clinical and operational use cases, **increasing model accuracy, safety, and contextual reliability by 30%**.
- Engineered **AI-powered workflow automation and decision-support prototypes** using Python, SQL, BigQuery, and Vertex AI, transforming operational data into scalable insights and **reducing manual analysis effort by 40%**.
- Designed evaluation frameworks for **50+ LLM test scenarios** covering prompt testing, error analysis, edge-case validation, and response quality checks to strengthen reliability across regulated, stakeholder-facing AI systems.
- Developed stakeholder-facing dashboards and KPI reports to monitor **5+ product adoption metrics** including login trends, feature usage, funnel conversion, and engagement, helping product owners identify drop-off points and prioritize roadmap improvements.
- Strengthened AI product readiness by identifying workflow bottlenecks, adversarial risks, and edge cases, converting findings into actionable user stories and **increasing solution robustness by 40%**.

Software Engineer | Gradient

Dec 2024 – Feb 2025

Product Analytics, A/B Testing, Email Automation, Campaign Analysis, User Engagement

Remote, USA

- Automated email outreach campaigns with **A/B testing, audience segmentation, and performance analysis**, reducing manual outreach effort by **80%** and informing data-driven user acquisition strategies.
- Analyzed **15+ mobile app feature and engagement metrics** to identify product improvement opportunities, supporting roadmap prioritization across user behavior, retention, and product usage trends.

Machine Learning Engineer | OneChimera Incorporated

May 2024 – Dec 2024

AI Experimentation, Data Pipelines, LangChain Automation, Diffusion Models, Model Evaluation

Remote, USA

- Led AI experimentation and **mentored 2 machine learning interns** on model evaluation, prompt testing, documentation standards, and analytical rigor, improving consistency across research workflows.
- Architected scalable data collection and processing pipelines to scrape, clean, and structure **100,000+ architecture images**, creating high-quality datasets for model training and downstream analytics.
- Integrated AI frameworks including **LangChain** to automate research workflows, scale data pipeline execution, and **reduce project turnaround time by 40%**.

PROJECTS

OK-VQA Using Multimodal RAG | Python, Deep Learning, NLP, LLMs, Generative AI.

- Developed a **Retrieval-Augmented Generation** pipeline using **BLIP-2** for Outside Knowledge Visual Question Answering, reducing dependence on full model retraining while strengthening answer grounding.
- Constructed a multimodal retrieval system using **CLIP Text Encoder, Hugging Face AutoTokenizer, and FAISS indexing** to retrieve external knowledge across **1,000+** visual question-answering examples.
- Incorporated external Wiki knowledge dynamically during inference, **resulting in a 20% increase in zero-shot accuracy**.

University Recommendation System | Python, EDA, Web Scraping, Data Mining, Predictive Analytics.

- Directed a **4-member team** to design and build a university recommendation system, **reducing student research time by 50% and improving engagement and enrollment intent by 25%**.
- Crafted a machine learning algorithm using **Selenium** and **Weighted KNN** to analyze student data and recommend suitable universities based on the student's academic background, test scores, and other preferences.
- Created a user-friendly interface for the recommendation system using **Bootstrap** and **HTML** for Frontend and **Flask** for Backend, **resulting in a 30% increase in user engagement**.

TECHNICAL SKILLS

Programming & ML: Python, PySpark, FastAPI, Go, R, Java, React, Git, GitHub

Data Science & Machine Learning: Statistics, Hypothesis Testing, Regression, Classification, Clustering, Predictive Modeling, Feature Engineering, A/B Testing, Experimentation, XGBoost, NLP, LLMs, Prompt Engineering, Model Evaluation

Data & Analytics Engineering: SQL, BigQuery, PostgreSQL, MySQL, MongoDB, Apache Spark, Spark SQL, ETL/ELT Pipelines

Cloud & DevOps: GCP, AWS, Azure, Docker, Kubernetes, Dataproc, Azure DevOps, JIRA, Confluence, Agile

BI & Product Analytics: Tableau, Power BI, Looker Studio, KPI Frameworks, Product Analytics, Funnel Analysis, User Engagement Analytics, Stakeholder Reporting, Executive Dashboards, Unit Economics

EDUCATION

Master of Science, Data Science

Indiana University Bloomington, Bloomington, Indiana

Aug 2022 – May 2024

GPA: 3.68/4.00

Bachelor of Engineering, Computer Engineering

K.J. Somaiya Institute of Technology, Mumbai, India

Aug 2015 – May 2019

GPA: 3.7/4.00